

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407908
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Vitale; Nicholas et al.

Heatsink of an image capture device

Abstract

An image capture device having a housing and a heatsink. The image capture device includes an integrated sensor and lens assembly. The heatsink is located partially or completely within the housing. The heatsink includes a through which a portion of the integrated sensor and lens assembly extends, and a planar surface located adjacent to the cutout. The planar surface includes a front side and a rear side. A printed circuit board in communication with a rear side of the planar surface of the heatsink. An LCD recess located in a front side of the planar surface of the heatsink that is configured to receive a liquid crystal display, wherein the LCD recess is opposite the printed circuit board.

Inventors: Vitale; Nicholas (Foster City, CA), Vargas Gonzalez; Raul (Redwood City, CA), Wong; Herman (Millbrae, CA)

Applicant: GoPro, Inc. (San Mateo, CA)

Family ID: 77464380

Assignee: GoPro, Inc. (San Mateo, CA)

Appl. No.: 18/210817

Filed: June 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230328345 A1	Oct. 12, 2023

Related U.S. Application Data

continuation parent-doc US 17378992 20210719 US 11689790 child-doc US 18210817
continuation-in-part parent-doc US 16803139 20200227 US 11277545 20220315 child-doc US 17378992

Publication Classification

Int. Cl.: **H04N23/51** (20230101); **G03B17/12** (20210101); **G03B17/56** (20210101); **H04N23/54** (20230101); **H04N23/57** (20230101); **H05K1/02** (20060101); **H05K1/18** (20060101); **G03B17/55** (20210101)

U.S. Cl.:

CPC **H04N23/51** (20230101); **G03B17/12** (20130101); **G03B17/561** (20130101); **H04N23/54** (20230101); **H04N23/57** (20230101); **H05K1/021** (20130101); **H05K1/18** (20130101); **G03B17/55** (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5471027	12/1994	Call	N/A	N/A
6097598	12/1999	Miyahara	N/A	N/A
7329869	12/2007	Cassel	N/A	N/A
9025080	12/2014	Samuels	N/A	N/A
9860970	12/2017	Tolbert	N/A	N/A
9910342	12/2017	Samuels	N/A	N/A
10613420	12/2019	Ali	N/A	N/A
10656689	12/2019	Kilgore	N/A	N/A
10701249	12/2019	Guo	N/A	N/A
11146711	12/2020	Crow	N/A	N/A
11277545	12/2021	Crow	N/A	N/A
11637949	12/2022	Crow	N/A	N/A
2003/0128536	12/2002	Radu	N/A	N/A
2003/0202328	12/2002	Deeney	N/A	N/A
2004/0169771	12/2003	Washington	N/A	N/A
2006/0100336	12/2005	Fukui	N/A	N/A
2008/0056695	12/2007	Huang	N/A	N/A
2008/0205881	12/2007	Sakurai	N/A	N/A
2009/0274953	12/2008	Myers	N/A	N/A
2010/0296805	12/2009	Mayer	N/A	N/A
2011/0127912	12/2010	Lee	N/A	N/A
2011/0194009	12/2010	Park	348/E5.022	H04N 23/54
2012/0035418	12/2011	Talbert	N/A	N/A
2014/0055671	12/2013	Kawamura	348/374	H04N 5/64
2014/0104479	12/2013	Samuels	N/A	N/A
2014/0160284	12/2013	Achenbach	N/A	N/A
2015/0049243	12/2014	Samuels	N/A	N/A
2015/0365569	12/2014	Mai	N/A	N/A
2016/0174408	12/2015	Tolbert	N/A	N/A

2016/0334692	12/2015	Samuels	N/A	N/A
2016/0349601	12/2015	Kungl	N/A	G03B 17/14
2016/0374190	12/2015	Tolbert	N/A	N/A
2017/0070652	12/2016	Müller	N/A	N/A
2017/0156205	12/2016	Liu	N/A	N/A
2017/0195531	12/2016	Warren	N/A	N/A
2017/0195533	12/2016	Seo	N/A	N/A
2017/0200992	12/2016	Piggott	N/A	N/A
2017/0294694	12/2016	Tso	N/A	N/A
2018/0107099	12/2017	Yasuda	N/A	N/A
2018/0259831	12/2017	Chiu	N/A	N/A
2019/0041600	12/2018	Sakamoto	N/A	N/A
2019/0056643	12/2018	Chang	N/A	N/A
2019/0163037	12/2018	Takashi	N/A	N/A
2019/0346126	12/2018	Wada	N/A	N/A
2020/0033774	12/2019	Shinji	N/A	N/A
2020/0064533	12/2019	Miyazaki	N/A	N/A
2020/0344395	12/2019	Guo	N/A	H05K 7/20309
2020/0413567	12/2019	Chusseau	N/A	N/A
2021/0033815	12/2020	Lin	N/A	N/A
2021/0122299	12/2020	Garcia	N/A	N/A
2021/0272879	12/2020	Kubo	N/A	H01L 23/3733
2021/0274067	12/2020	Crow	N/A	N/A
2021/0306536	12/2020	Vitale	N/A	N/A
2021/0321027	12/2020	Crow	N/A	N/A
2021/0344819	12/2020	Vitale	N/A	N/A
2022/0021795	12/2021	Crow	N/A	N/A
2022/0124226	12/2021	Jeong	N/A	N/A
2022/0159148	12/2021	Crow	N/A	N/A
2023/0009451	12/2022	Connolly	N/A	N/A
2024/0073499	12/2023	Vitale	N/A	N/A
2024/0073500	12/2023	Vitale	N/A	N/A
2024/0073506	12/2023	Vitale	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110235059	12/2018	CN	N/A
2004020798	12/2003	JP	N/A
2006086752	12/2005	JP	N/A
2008015476	12/2007	JP	N/A
2009278584	12/2008	JP	N/A
5322866	12/2012	JP	N/A
2017073634	12/2016	JP	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability issued in App. No. PCT/US2021/019970, mailing date Sep. 9, 2022, 6 pages. cited by applicant

International Preliminary Report on Patentability issued in App. No. PCT/US2021/026140, mailing

date Oct. 20, 2022, 8 pages. cited by applicant

International Search Report and Written Opinion for App. No. PCT/US2021/019970, dated Jun. 3, 2021, 9 pages. cited by applicant

International Search Report and Written Opinion for App. No. PCT/US2021/026140, dated Jul. 1, 2021, 10 pages. cited by applicant

U.S. Appl. No. 16/803,139, filed Feb. 27, 2020, Crow et al. cited by applicant

Primary Examiner: Haskins; Twyler L

Assistant Examiner: Chiu; Wesley J

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/378,992, filed on Jul. 19, 2021, which is a continuation in part of U.S. patent application Ser. No. 16/803,139, filed on Feb. 27, 2020, now U.S. Pat. No. 11,277,545, the entire disclosures of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) This disclosure relates to a heatsink in an image capture device and specifically a heatsink for an image capture device.

BACKGROUND

(2) Photography during physical activity has been improved by use of simple-to-operate, lightweight, compact cameras or imaging devices. Advancements in technology have allowed for more functionality to be added to the cameras. Faster components have been added to the cameras, allowing for clearer images to be taken by the cameras. Additionally, the cameras have been sealed so that the cameras may be splashed, submerged, or otherwise inundated with water. These improvements have caused an increase in thermal loads within the camera.

SUMMARY

(3) Disclosed herein are implementations of a heatsink for an image capture device.

(4) The present teachings provide an image capture device comprising a housing, a heatsink, a printed circuit board, and a sheet conductor. The heatsink is located partially or completely within the housing, wherein the heatsink comprises a planar surface. The printed circuit board is in communication with the heatsink. The sheet conductor is a graphite sheet connected to the planar surface of the heatsink and in direct contact with the printed circuit board or components of the printed circuit board.

(5) The present teachings provide an image capture device comprising: a heatsink and a sheet conductor. The heatsink comprises a planar surface and a cut out that are adjacent. The sheet conductor covers a portion of the planar surface of the heatsink. The sheet conductor has a thickness that is less than the thickness of the heatsink and the sheet conductor is made of a material that has a higher thermal conductivity than the material of the heatsink.

(6) The present teachings provide an image capture device comprising a housing, a heatsink, a printed circuit board, a sheet conductor, and a connector. The heatsink is located partially or completely within the housing, wherein the heatsink comprises a planar surface. The printed circuit board is in communication with the heatsink. The sheet conductor is located between and in direct contact with the planar surface of the heatsink and the printed circuit board. The connector connects the sheet conductor to the heatsink.

(7) An image capture device having a housing and a heatsink. The image capture device includes an integrated sensor and lens assembly. The heatsink is located partially or completely within the housing. The heatsink includes a through which a portion of the integrated sensor and lens assembly extends, and a planar surface located adjacent to the cutout. The planar surface includes a front side and a rear side. A printed circuit board in communication with a rear side of the planar surface of the heatsink. An LCD recess located in a front side of the planar surface of the heatsink that is configured to receive a liquid crystal display, wherein the LCD recess is opposite the printed circuit board.

(8) The present teachings provide: an image capture device including: an integrated sensor and lens assembly and a heatsink. The heatsink includes a planar surface and a cutout. The planar surface includes a first side facing external to the image capture device and a second side facing internal to the image capture device. The cutout is adjacent to the planar surface, wherein a portion of the integrated sensor and lens assembly is located on a first side of the heatsink, the integrated sensor and lens assembly extends through the cutout, and a portion of the integrated sensor and lens assembly is located on a second side of the heatsink. The heatsink includes bosses configured to receive fasteners that form a connection with the heatsink; and an LCD recess, on the first side of the planar surface, that is configured to connect to a liquid crystal display (LCD) of the image capture device. A printed circuit board in communication with a second side of the planar surface.

(9) The present teachings provide: an image capture device including a housing, a heatsink, a thermal interface material, and a printed circuit board. The heatsink located partially or completely within the housing. The heatsink includes: a planar surface, a cutout, a first mounting flange connected to a first edge of the heatsink, and a finger mounting flange connected to a second edge of the heatsink with the planar surface extending between the first mounting flange and the finger mounting flange. The printed circuit board in communication with the planar surface of the heatsink via the thermal interface material.

(10) Additional teachings are described in further detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) FIGS. 1A-B are isometric views of an example of an image capture device.

(3) FIGS. 2A-B are isometric views of another example of an image capture device.

(4) FIG. 3 is a block diagram of electronic components of an image capture device.

(5) FIG. 4A is a front perspective view of a heatsink, an ISLA, and a mounting structure of an image capture device.

(6) FIG. 4B is a partial exploded view of the image capture device of FIG. 1A.

(7) FIG. 4C is a rear view of the ISLA connected to the mounting structure and extending through the heatsink of FIGS. 4A-4B.

(8) FIG. 4D is a side view of the mounting structure, the housing, the heatsink, and the ISLA of FIGS. 4A-4C.

(9) FIG. 5 is a rear perspective view of a battery cage connected to a heatsink.

(10) FIG. 6A is a rear perspective view of the heatsink of FIGS. 4A-4D.

(11) FIG. 6B is a front view of the heatsink of FIG. 6A.

(12) FIG. 6C is a rear view of the heatsink of FIGS. 6A-6B.

(13) FIG. 7A is a partial top perspective view of an example of a global positioning system (GPS)

printed circuit board and the heatsink of FIGS. 4A-4D.

(14) FIG. 7B is a partial cross-sectional view of an image capture device showing the GPS printed circuit board of FIG. 7A extending below the housing.

(15) FIG. 8A is a rear isometric view of a heatsink including a sheet conductor.

(16) FIG. 8B is a rear isometric view of the heatsink and the sheet conductor of FIG. 8A covered by a printed circuit board.

(17) FIG. 9 is a front view of the printed circuit board of FIG. 8B.

(18) FIG. 10A is a cross-sectional view of the heatsink, sheet conductor, and printed circuit board of FIG. 8B along line XA-XA.

(19) FIG. 10B is a cross-sectional view illustrating an interface with the sheet conductor of FIG. 8B along line XB-XB.

(20) FIG. 10C is an exploded view of the components of FIGS. 10A-10B.

DETAILED DESCRIPTION

(21) As more components and processing power are added to image capture devices, more heat is added to image capture devices. Improved thermal management in an image capture device becomes important to provide for extended use of the image capture device. Management of heat with and along a heatsink may allow for extended use of the image capture device before the heat retained within the image capture device impacts operation or before components become potentially damaged. The heatsink may have a hole that an (ISLA) extends through, where the ISLA is free of contact with the heatsink. The heatsink may support a battery and/or a battery cage, may support a sensor, circuit board, or other component in a GPS, or may include a combination of the described components. The present disclosure relates to heatsinks in image capture devices such as heatsinks that are free of communication with an ISLA, create a space between a housing and components of a GPS, support a battery cage, or include a combination of these components.

(22) FIGS. 1A-B are isometric views of an example of an image capture device **100**. The image capture device **100** may include a body **102**, a lens **104** structured on a front surface of the body **102**, various indicators on the front surface of the body **102** (such as light-emitting diodes (LEDs), displays, and the like), various input mechanisms (such as buttons, switches, and/or touch-screens), and electronics (such as imaging electronics, power electronics, etc.) internal to the body **102** for capturing images via the lens **104** and/or performing other functions. The lens **104** is configured to receive light incident upon the lens **104** and to direct received light onto an image sensor internal to the body **102**. The image capture device **100** may be configured to capture images and video and to store captured images and video for subsequent display or playback.

(23) The image capture device **100** may include an LED or another form of indicator **106** to indicate a status of the image capture device **100** and a liquid-crystal display (LCD) or other form of a display **108** to show status information such as battery life, camera mode, elapsed time, and the like. The image capture device **100** may also include a mode button **110** and a shutter button **112** that are configured to allow a user of the image capture device **100** to interact with the image capture device **100**. For example, the mode button **110** and the shutter button **112** may be used to turn the image capture device **100** on and off, scroll through modes and settings, and select modes and change settings. The image capture device **100** may include additional buttons or interfaces (not shown) to support and/or control additional functionality.

(24) The image capture device **100** may include a door **114** coupled to the body **102**, for example, using a hinge mechanism **116**. The door **114** may be secured to the body **102** using a latch mechanism **118** that releasably engages the body **102** at a position generally opposite the hinge mechanism **116**. The door **114** may also include a seal **120** and a battery interface **122**. When the door **114** is an open position, access is provided to an input-output (I/O) interface **124** for connecting to or communicating with external devices as described below and to a battery receptacle **126** for placement and replacement of a battery (not shown). The battery receptacle **126** includes operative connections (not shown) for power transfer between the battery and the image

capture device **100**. When the door **114** is in a closed position, the seal **120** engages a flange (not shown) or other interface to provide an environmental seal, and the battery interface **122** engages the battery to secure the battery in the battery receptacle **126**. The door **114** can also have a removed position (not shown) where the entire door **114** is separated from the image capture device **100**, that is, where both the hinge mechanism **116** and the latch mechanism **118** are decoupled from the body **102** to allow the door **114** to be removed from the image capture device **100**.

(25) The image capture device **100** may include a microphone **128** on a front surface and another microphone **130** on a side surface. The image capture device **100** may include other microphones on other surfaces (not shown). The microphones **128**, **130** may be configured to receive and record audio signals in conjunction with recording video or separate from recording of video. The image capture device **100** may include a speaker **132** on a bottom surface of the image capture device **100**. The image capture device **100** may include other speakers on other surfaces (not shown). The speaker **132** may be configured to play back recorded audio or emit sounds associated with notifications.

(26) A front surface of the image capture device **100** may include a drainage channel **134**. A bottom surface of the image capture device **100** may include an interconnect mechanism **136** for connecting the image capture device **100** to a handle grip or other securing device. In the example shown in FIG. **1B**, the interconnect mechanism **136** includes folding protrusions configured to move between a nested or collapsed position as shown and an extended or open position (not shown) that facilitates coupling of the protrusions to mating protrusions of other devices such as handle grips, mounts, clips, or like devices.

(27) The image capture device **100** may include an interactive display **138** that allows for interaction with the image capture device **100** while simultaneously displaying information on a surface of the image capture device **100**.

(28) The image capture device **100** of FIGS. **1A-B** includes an exterior that encompasses and protects internal electronics. In the present example, the exterior includes six surfaces (i.e. a front face, a left face, a right face, a back face, a top face, and a bottom face) that form a rectangular cuboid. Furthermore, both the front and rear surfaces of the image capture device **100** are rectangular. In other embodiments, the exterior may have a different shape. The image capture device **100** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass. The image capture device **100** may include features other than those described here. For example, the image capture device **100** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **100**.

(29) The image capture device **100** may include various types of image sensors, such as charge-coupled device (CCD) sensors, active pixel sensors (APS), complementary metal-oxide-semiconductor (CMOS) sensors, N-type metal-oxide-semiconductor (NMOS) sensors, and/or any other image sensor or combination of image sensors.

(30) Although not illustrated, in various embodiments, the image capture device **100** may include other additional electrical components (e.g., an image processor, camera system-on-chip (SoC), etc.), which may be included on one or more circuit boards within the body **102** of the image capture device **100**.

(31) The image capture device **100** may interface with or communicate with an external device, such as an external user interface device (not shown), via a wired or wireless computing communication link (e.g., the I/O interface **124**). Any number of computing communication links may be used. The computing communication link may be a direct computing communication link or an indirect computing communication link, such as a link including another device or a network, such as the internet, may be used.

(32) In some implementations, the computing communication link may be a Wi-Fi link, an infrared link, a Bluetooth (BT) link, a cellular link, a ZigBee link, a near field communications (NFC) link,

such as an ISO/IEC 20643 protocol link, an Advanced Network Technology interoperability (ANT+) link, and/or any other wireless communications link or combination of links.

(33) In some implementations, the computing communication link may be an HDMI link, a USB link, a digital video interface link, a display port interface link, such as a Video Electronics Standards Association (VESA) digital display interface link, an Ethernet link, a Thunderbolt link, and/or other wired computing communication link.

(34) The image capture device **100** may transmit images, such as panoramic images, or portions thereof, to the external user interface device via the computing communication link, and the external user interface device may store, process, display, or a combination thereof the panoramic images.

(35) The external user interface device may be a computing device, such as a smartphone, a tablet computer, a phablet, a smart watch, a portable computer, personal computing device, and/or another device or combination of devices configured to receive user input, communicate information with the image capture device **100** via the computing communication link, or receive user input and communicate information with the image capture device **100** via the computing communication link.

(36) The external user interface device may display, or otherwise present, content, such as images or video, acquired by the image capture device **100**. For example, a display of the external user interface device may be a viewport into the three-dimensional space represented by the panoramic images or video captured or created by the image capture device **100**.

(37) The external user interface device may communicate information, such as metadata, to the image capture device **100**. For example, the external user interface device may send orientation information of the external user interface device with respect to a defined coordinate system to the image capture device **100**, such that the image capture device **100** may determine an orientation of the external user interface device relative to the image capture device **100**.

(38) Based on the determined orientation, the image capture device **100** may identify a portion of the panoramic images or video captured by the image capture device **100** for the image capture device **100** to send to the external user interface device for presentation as the viewport. In some implementations, based on the determined orientation, the image capture device **100** may determine the location of the external user interface device and/or the dimensions for viewing of a portion of the panoramic images or video.

(39) The external user interface device may implement or execute one or more applications to manage or control the image capture device **100**. For example, the external user interface device may include an application for controlling camera configuration, video acquisition, video display, or any other configurable or controllable aspect of the image capture device **100**.

(40) The user interface device, such as via an application, may generate and share, such as via a cloud-based or social media service, one or more images, or short video clips, such as in response to user input. In some implementations, the external user interface device, such as via an application, may remotely control the image capture device **100** such as in response to user input.

(41) The external user interface device, such as via an application, may display unprocessed or minimally processed images or video captured by the image capture device **100** contemporaneously with capturing the images or video by the image capture device **100**, such as for shot framing or live preview, and which may be performed in response to user input. In some implementations, the external user interface device, such as via an application, may mark one or more key moments contemporaneously with capturing the images or video by the image capture device **100**, such as with a tag or highlight in response to a user input or user gesture.

(42) The external user interface device, such as via an application, may display or otherwise present marks or tags associated with images or video, such as in response to user input. For example, marks may be presented in a camera roll application for location review and/or playback of video highlights.

(43) The external user interface device, such as via an application, may wirelessly control camera software, hardware, or both. For example, the external user interface device may include a web-based graphical interface accessible by a user for selecting a live or previously recorded video stream from the image capture device **100** for display on the external user interface device.

(44) The external user interface device may receive information indicating a user setting, such as an image resolution setting (e.g., 3840 pixels by 2160 pixels), a frame rate setting (e.g., 60 frames per second (fps)), a location setting, and/or a context setting, which may indicate an activity, such as mountain biking, in response to user input, and may communicate the settings, or related information, to the image capture device **100**.

(45) FIGS. 2A-B illustrate another example of an image capture device **200**. The image capture device **200** includes a body **202** and two camera lenses **204** and **206** disposed on opposing surfaces of the body **202**, for example, in a back-to-back configuration, Janus configuration, or offset Janus configuration. The body **202** of the image capture device **200** may be made of a rigid material such as plastic, aluminum, steel, or fiberglass.

(46) The image capture device **200** includes various indicators on the front of the surface of the body **202** (such as LEDs, displays, and the like), various input mechanisms (such as buttons, switches, and touch-screen mechanisms), and electronics (e.g., imaging electronics, power electronics, etc.) internal to the body **202** that are configured to support image capture via the two camera lenses **204** and **206** and/or perform other imaging functions.

(47) The image capture device **200** includes various indicators, for example, LEDs **208**, **210** to indicate a status of the image capture device **100**. The image capture device **200** may include a mode button **212** and a shutter button **214** configured to allow a user of the image capture device **200** to interact with the image capture device **200**, to turn the image capture device **200** on, and to otherwise configure the operating mode of the image capture device **200**. It should be appreciated, however, that, in alternate embodiments, the image capture device **200** may include additional buttons or inputs to support and/or control additional functionality.

(48) The image capture device **200** may include an interconnect mechanism **216** for connecting the image capture device **200** to a handle grip or other securing device. In the example shown in FIGS. 2A and 2B, the interconnect mechanism **216** includes folding protrusions configured to move between a nested or collapsed position (not shown) and an extended or open position as shown that facilitates coupling of the protrusions to mating protrusions of other devices such as handle grips, mounts, clips, or like devices.

(49) The image capture device **200** may include audio components **218**, **220**, **222** such as microphones configured to receive and record audio signals (e.g., voice or other audio commands) in conjunction with recording video. The audio component **218**, **220**, **222** can also be configured to play back audio signals or provide notifications or alerts, for example, using speakers. Placement of the audio components **218**, **220**, **222** may be on one or more of several surfaces of the image capture device **200**. In the example of FIGS. 2A and 2B, the image capture device **200** includes three audio components **218**, **220**, **222**, with the audio component **218** on a front surface, the audio component **220** on a side surface, and the audio component **222** on a back surface of the image capture device **200**. Other numbers and configurations for the audio components are also possible.

(50) The image capture device **200** may include an interactive display **224** that allows for interaction with the image capture device **200** while simultaneously displaying information on a surface of the image capture device **200**. The interactive display **224** may include an I/O interface, receive touch inputs, display image information during video capture, and/or provide status information to a user. The status information provided by the interactive display **224** may include battery power level, memory card capacity, time elapsed for a recorded video, etc.

(51) The image capture device **200** may include a release mechanism **225** that receives a user input to in order to change a position of a door (not shown) of the image capture device **200**. The release mechanism **225** may be used to open the door (not shown) in order to access a battery, a battery

receptacle, an I/O interface, a memory card interface, etc. (not shown) that are similar to components described in respect to the image capture device **100** of FIGS. **1A** and **1B**.

(52) In some embodiments, the image capture device **200** described herein includes features other than those described. For example, instead of the I/O interface and the interactive display **224**, the image capture device **200** may include additional interfaces or different interface features. For example, the image capture device **200** may include additional buttons or different interface features, such as interchangeable lenses, cold shoes, and hot shoes that can add functional features to the image capture device **200**.

(53) FIG. **3** is a block diagram of electronic components in an image capture device **300**. The image capture device **300** may be a single-lens image capture device, a multi-lens image capture device, or variations thereof, including an image capture device with multiple capabilities such as use of interchangeable ISLA. The description of the image capture device **300** is also applicable to the image capture devices **100**, **200** of FIGS. **1A-B** and **2A-B**.

(54) The image capture device **300** includes a body **302** which includes electronic components such as capture components **310**, a processing apparatus **320**, data interface components **330**, movement sensors **340**, power components **350**, and/or user interface components **360**.

(55) The capture components **310** include one or more image sensors **312** for capturing images and one or more microphones **314** for capturing audio.

(56) The image sensor(s) **312** is configured to detect light of a certain spectrum (e.g., the visible spectrum or the infrared spectrum) and convey information constituting an image as electrical signals (e.g., analog or digital signals). The image sensor(s) **312** detects light incident through a lens coupled or connected to the body **302**. The image sensor(s) **312** may be any suitable type of image sensor, such as a charge-coupled device (CCD) sensor, active pixel sensor (APS), complementary metal-oxide-semiconductor (CMOS) sensor, N-type metal-oxide-semiconductor (NMOS) sensor, and/or any other image sensor or combination of image sensors. Image signals from the image sensor(s) **312** may be passed to other electronic components of the image capture device **300** via a bus **380**, such as to the processing apparatus **320**. In some implementations, the image sensor(s) **312** includes a digital-to-analog converter. A multi-lens variation of the image capture device **300** can include multiple image sensors **312**.

(57) The microphone(s) **314** is configured to detect sound, which may be recorded in conjunction with capturing images to form a video. The microphone(s) **314** may also detect sound in order to receive audible commands to control the image capture device **300**.

(58) The processing apparatus **320** may be configured to perform image signal processing (e.g., filtering, tone mapping, stitching, and/or encoding) to generate output images based on image data from the image sensor(s) **312**. The processing apparatus **320** may include one or more processors having single or multiple processing cores. In some implementations, the processing apparatus **320** may include an application specific integrated circuit (ASIC). For example, the processing apparatus **320** may include a custom image signal processor. The processing apparatus **320** may exchange data (e.g., image data) with other components of the image capture device **300**, such as the image sensor(s) **312**, via the bus **380**.

(59) The processing apparatus **320** may include memory, such as a random-access memory (RAM) device, flash memory, or another suitable type of storage device, such as a non-transitory computer-readable memory. The memory of the processing apparatus **320** may include executable instructions and data that can be accessed by one or more processors of the processing apparatus **320**. For example, the processing apparatus **320** may include one or more dynamic random-access memory (DRAM) modules, such as double data rate synchronous dynamic random-access memory (DDR SDRAM). In some implementations, the processing apparatus **320** may include a digital signal processor (DSP). More than one processing apparatus may also be present or associated with the image capture device **300**.

(60) The data interface components **330** enable communication between the image capture device

300 and other electronic devices, such as a remote control, a smartphone, a tablet computer, a laptop computer, a desktop computer, or a storage device. For example, the data interface components **330** may be used to receive commands to operate the image capture device **300**, transfer image data to other electronic devices, and/or transfer other signals or information to and from the image capture device **300**. The data interface components **330** may be configured for wired and/or wireless communication. For example, the data interface components **330** may include an I/O interface **332** that provides wired communication for the image capture device, which may be a USB interface (e.g., USB type-C), a high-definition multimedia interface (HDMI), or a FireWire interface. The data interface components **330** may include a wireless data interface **334** that provides wireless communication for the image capture device **300**, such as a Bluetooth interface, a ZigBee interface, and/or a Wi-Fi interface. The data interface components **330** may include a storage interface **336**, such as a memory card slot configured to receive and operatively couple to a storage device (e.g., a memory card) for data transfer with the image capture device **300** (e.g., for storing captured images and/or recorded audio and video).

(61) The movement sensors **340** may detect the position and movement of the image capture device **300**. The movement sensors **340** may include a position sensor **342**, an accelerometer **344**, or a gyroscope **346**. The position sensor **342**, such as a GPS sensor, is used to determine a position of the image capture device **300**. The accelerometer **344**, such as a three-axis accelerometer, measures linear motion (e.g., linear acceleration) of the image capture device **300**. The gyroscope **346**, such as a three-axis gyroscope, measures rotational motion (e.g., rate of rotation) of the image capture device **300**. Other types of movement sensors **340** may also be present or associated with the image capture device **300**.

(62) The power components **350** may receive, store, and/or provide power for operating the image capture device **300**. The power components **350** may include a battery interface **352** and a battery **354**. The battery interface **352** operatively couples to the battery **354**, for example, with conductive contacts to transfer power from the battery **354** to the other electronic components of the image capture device **300**. The power components **350** may also include the I/O interface **332**, as indicated in dotted line, and the power components **350** may receive power from an external source, such as a wall plug or external battery, for operating the image capture device **300** and/or charging the battery **354** of the image capture device **300**.

(63) The user interface components **360** may allow the user to interact with the image capture device **300**, for example, providing outputs to the user and receiving inputs from the user. The user interface components **360** may include visual output components **362** to visually communicate information and/or present captured images to the user. The visual output components **362** may include one or more lights **364** and/or more displays **366**. The display(s) **366** may be configured as a touch screen that receives inputs from the user. The user interface components **360** may also include one or more speakers **368**. The speaker(s) **368** can function as an audio output component that audibly communicates information and/or presents recorded audio to the user. The user interface components **360** may also include one or more physical input interfaces **370** that are physically manipulated by the user to provide input to the image capture device **300**. The physical input interfaces **370** may, for example, be configured as buttons, toggles, or switches. The user interface components **360** may also be considered to include the microphone(s) **314**, as indicated in dotted line, and the microphone(s) **314** may function to receive audio inputs from the user, such as voice commands.

(64) FIGS. 4A-D are views of internal components of an image capture device such as the image capture devices **100**, **200** of FIGS. 1A-2B. The views depict the heatsink **400** having mounting flanges **402**, component recesses **404A** and **404B**, and a cutout **412**.

(65) FIG. 4A is a front perspective view of a heatsink **400** having component recesses **404A**, B. As shown, a first component recess **404A** is an LCD recess **406** that may receive an LCD screen (not shown) and a second component recess **404B** is a microphone membrane recess **408** that may

receive a microphone, a microphone membrane, or both (not shown). A bayonet **426** extends on an outer side of the heatsink **400** and housing (not shown) so that additional lenses or lens covers (not shown) may be connected to the image capture device proximate to or covering an ISLA **430**. A bayonet **426** as discussed herein is an example of one type or style of mounting structure that may be used to connect the ISLA **430**, a cover lens (not shown), or both. Other types of mounting structures, such as mounting structures having threads, detents, slots, or other connection features are also possible. The bayonet **426** is connected to a housing by fasteners **440** extending from a forward side of the bayonet **426** towards a rear side of the bayonet **426**. The ISLA **430** extends through the heatsink **400** and into the bayonet **426** so that a forward end **450** of the ISLA **430** is connected and stabilized by fasteners **446** (FIG. **4B**) extending from the rear side of the bayonet **426** towards the forward side of the bayonet **426**. The forward end **450** of the ISLA **430** is connected to the bayonet **426**, which acts as a stabilization point for the ISLA **430** and combats movement of the optical axis.

(66) The bayonet **426** is located external to the housing (not shown) and the heatsink **400** so that the ISLA **430** is connected in a forward region of the image capture device to stabilize the ISLA **430**. The bayonet **426** may function to support the forward end **450** of the ISLA **430**. A forward end of the ISLA **430** may extend beyond the bayonet **426**. A forwardmost end of the ISLA **430** may be flush with a forward surface of the bayonet **426**. A forwardmost end of the ISLA **430** may be counter sunk into the bayonet **426**. The ISLA **430** may be free of direct contact with the housing (not shown), the heatsink **400**, or both. The forward end **450** of the ISLA **430** may be supported only by the bayonet **426**. The forward end **450**, a rear end (not shown), or both ends of the ISLA **430** may be connected to the bayonet **426**, the housing, the heatsink **400**, or a combination thereof.

(67) FIG. **4B** is a partial exploded view of an image capture device. The exploded view shows the heatsink **400** of FIG. **4A** separated from the bayonet **426**, the ISLA **430**, and a housing **436**. The ISLA **430** extends through the cutout **412** so that the ISLA **430** can directly connect to the bayonet **426** via the fasteners **446**. The heatsink **400** and the housing **436** are connected via a combination of fasteners **442** (FIG. **4C**) and heat stakes (not shown) so that the image capture device has supported internal components. The bayonet **426** is connected to an opposing surface of the housing **436** as the heatsink **400** by fasteners **444** so that all of the components are grounded to the housing **436** directly or indirectly. For example, the bayonet **426** and heatsink **400** may be grounded through a conductive gasket formed, for example, of silicone. In another example, the bayonet **426** may be directly connected to both the heatsink **400** and the housing **436**. The bayonet **426** may only be connected to the housing **436**. The bayonet **426** may be located on an external surface of the housing **436**. The bayonet **426** and the heatsink **400** may be spaced apart by the housing **436** so that the bayonet is located further forward within the image capture device than the heatsink **400**. In FIG. **4B**, the housing **436** extends around the bayonet **426** such that a portion of the bayonet **426** is still exposed to receive different lenses, filters, or lens covers (not shown). For example, the housing **436** may cover the fasteners **444** and leave connection features **428** exposed.

(68) FIG. **4C** illustrates a rear view depicting the heatsink **400** of FIGS. **4A-4B**. The bayonet **426** is located on a forward side of the heatsink **400** so that the ISLA **430** extends from a rear side of the heatsink **400** through the cutout **412** and into contact with the bayonet **426**. The heatsink **400** is connected to and supported by a housing (not shown) by fasteners **442** and heat stakes (not shown). The ISLA **430** includes mounting arms **432** that receive fasteners **446** to connect the ISLA to the bayonet **426**. The cutout **412** may be shaped to mirror the shape of the ISLA **430**, the mounting arms **432**, or both. The cutout **412** is a space through the heatsink that is surrounded by heatsink material. The cutout **412** and the ISLA **430** may be complementary in shape. The fasteners **446** extend through mounting arms **432** of the ISLA **430** into the bayonet **426**; however, as illustrated, one of the fasteners **446** has been removed from a boss **433** in one of the mounting arms **432** to illustrate a passage **438** through the mounting arm **432** into the bayonet **426**. As shown, the mounting arms **432** may be coplanar with the cutout **412**. The mounting arms **432** may each

include a boss **433**. The bosses **433** may receive the fasteners **446**. The bosses **433** may be circular, oval, U-shaped, or C shaped. The mounting arms **432** may function to hold a front of the ISLA **430** relative to the bayonet **426**.

(69) FIG. **4D** is a partial cross-sectional view illustrating the relationship of the heatsink **400**, the bayonet **426**, and the housing **436** of an image capture device when connected together via the fasteners **442**, **444**, **446**. The heatsink **400** and the housing **436** are connected via fasteners **442** (FIG. **4C**). The bayonet **426** is connected to the housing **436** by fasteners **444** extending from an external location with respect to the image capture device to an internal location. The bayonet **426** functions to support a forward end **450** of the ISLA **430** so that the ISLA **430** is free of movement relative to the housing **436**, the heatsink **400**, or both. The bayonet **426** is located proximate to a forward end **450** of the ISLA **430** (e.g., within a range between 1 mm and 7 mm or a range between 3 mm and 5 mm).

(70) As shown in FIG. **4D**, the bayonet **426** may function to align the ISLA **430** relative to the housing **436**, the heatsink **400**, or both so that images or videos may be captured by a user without the ISLA **430** moving relative to the housing **436**. The bayonet **426** may be partially or fully located outside of the housing **436**. The mounting arms **432** (FIG. **4C**), the ISLA **430**, or both may be free of contact with the housing **436**. The fasteners **442**, **444**, **446** may extend from an internal location (e.g., a rear side of the heatsink **400**, the housing **436**, the bayonet **426**, or a combination thereof) toward an external location of the image capture device. The ISLA **430** includes the forward end **450** that extends through the heatsink **400**, the housing **436**, and the bayonet **426**, and is connected to the bayonet via the fasteners **446**. A rear end **452** of the ISLA **430** extends into an internal location of the image capture device, for example, below a GPS assembly **454**. The GPS assembly **454** is connected to one of the mounting flanges **402** of the heatsink **400** as shown.

(71) FIG. **5** is a rear perspective view of a connection between a battery cage **502** and a heatsink **504**. The heatsink **504** includes mounting flanges **506**, **508** and a finger mounting flange **510** extending outward from the heatsink **504**. The finger mounting flange **510** receives fasteners **512** that connect a housing (not shown) to the finger mounting flange **510**. The finger mounting flange **510** functions to connect the heatsink **504** to a housing (not shown), connect the heatsink **504** to the battery cage **502**, or both. The finger mounting flange **510** functions to assist in forming a watertight connection between the housing (FIG. **4B**) and the heatsink **504**. The finger mounting flange **510** includes seal recesses **514**, seal members **516**, and a space **518** that is located between the seal recesses **514**. The finger mounting flange **510** may include the seal recesses **514**.

(72) In another example, the seal recesses **514** and the seal members **516** may be located in or on the housing. The seal recesses **514** function to partially or fully counter sink the seal members **516**. Each seal recess **514** may receive one of the seal members **516**. The finger mounting flange **510** may include a same number of seal members **516** and seal recesses **514**. The seal members **516** and the seal recesses **514** may be located on a bottom surface of the finger mounting flange **510**, between the finger mounting flange **510** and the housing (not shown), or both. For example, the finger mounting flange **510** may include two seal recesses **514** and the housing may include two seal recesses **514**. The seal recesses **514**, seal members **516**, or both may be circular, square, rectangular, triangular, geometric, symmetrical, non-symmetrical, or a combination thereof. The seal recesses **514** may be ring recesses. The seal members **516** may be located between the housing and the finger mounting flange **510**. The finger mounting flange **510** may have an area that receives one or more, two or more, three or more, or even four or more seal members **516**. The space **518** may be a reduction in material from the heatsink **504** where heat is not introduced into the heatsink **504**. The seal recesses **514** and seal members **516** may be located on an opposite side of the finger mounting flange **510** as the battery cage **502**.

(73) The finger mounting flange **510** is connected to the battery cage **502** by a battery connector **522**. The battery cage **502** is also connected to a planar surface **524** of the heatsink **504** via a fastener **526** that extend through a cage connector **542**. The battery cage **502** functions to transmit

heat between a battery (not shown) and the heatsink **504**, retain a battery within the image capture device, or both. The battery cage **502** may be in direct or indirect contact with the finger mounting flanges **510**, the planar surface **524**, or both. The battery cage **502** may be made of or include metal, plastic, a polymer, rubber, an elastomer, or a combination thereof. The battery cage **502** may be made of a metal comprising aluminum, titanium, steel, iron, or a combination thereof. The battery cage **502** may be solid or may include through holes **534**. The through holes **534** may be round, square, rectangular, symmetrical, asymmetrical geometric, non-geometric, or a combination thereof. The through holes **534** may be located within a forward side, a rearward side, or both sides of the battery cage **502**. A forward side of the battery cage **502** may be directly connected to a rear side of the heatsink **504**. The battery cage **502** and the heatsink **504** may be connected via the fasteners **512**. The battery cage **502** may have a planar surface (not shown), the heatsink **504** may have the planar surface **524**, and the two planar surfaces may be in contact with each other. The battery cage **502** may be directly or indirectly connected to the finger mounting flange **510**. The battery cage **502** may be connected to both the planar surface **524** and the finger mounting flange **510** of the heatsink **504**. A battery connector **522** may connect the battery cage **502** to the finger mounting flange **510**.

(74) The battery connector **522** extends between the finger mounting flange **510** and the battery cage **502**. The battery connector **522** functions to restrict movement of the battery cage **502** relative to the heatsink **504**, to maintain contact between the battery cage **502** and the heatsink **504**, or both. For example, the battery connector **522** may retain the battery cage **502** against the planar surface **524** of the heatsink **504**. The battery connector **522** may be generally “L” shaped, “C” shaped, straight, curved, or a combination thereof. The battery connector **522** may be a monolithic part of or a discrete part from the battery cage **502**, the finger mounting flange **510**, or both. The battery connector **522** may extend downward from the battery cage **502** over a battery boss **536** in the finger mounting flange **510** so that a fastener **526** may extend through and connect the battery cage **502** to the finger mounting flange **510**. The battery connector **522** may be flat, smooth, ribbed **540**, include raised surfaces, or a combination thereof. The battery connector **522** may be located opposite a GPS assembly **528**.

(75) The mounting flange **508** of the heatsink **504** is connected to and supports the GPS assembly **528** so that the GPS assembly **528** is fixedly connected within the image capture device. The GPS assembly **528** includes a printed circuit board (PCB) **530** and an antenna **532**. The antenna **532** is fixed near an outside of the image capture device by the mounting flange **508** of the heatsink **504** so that the antenna **532** can accurately send and receive information.

(76) FIGS. **6A-C** are various views of a heatsink **600**. The heatsink **600** can be used with an image capture device such as the image capture devices **100**, **200** of FIGS. **1A-2B** so that thermal energy within the image capture device is distributed and moved away from components that have thermal energy. The heatsink **600** includes mounting flanges **602A**, **602B**, and **602C** that extend outward from a body portion of the heatsink **600** to support components. The heatsink **600** includes component recesses **604** including an LCD recess **606**, a microphone membrane recess **608**, and a microphone recess **610** so that components located within the recesses **604**, **606**, **608**, **610** are at least partially housed within the heatsink **600**.

(77) FIG. **6A** is a rear perspective view of the heatsink **600**, which functions to remove heat from components, distribute heat, redistribute heat, prevent components from becoming overheated, or to perform a combination thereof. In one example, the heatsink **600** may have a portion that is located external to a housing (not shown) of the image capture device. In another example, the heatsink **600** may have a portion that is located internal to a housing (not shown) of the image capture device. The heatsink **600** may be located entirely internal to a housing, for example, if used with the image capture devices **100**, **200** of FIGS. **1A-2B**. The heatsink **600** may be located entirely external to a housing (not shown). The heatsink **600** may include a body portion. The body portion of the heatsink **600** may include a planar surface that extends parallel to a forward surface, a

rearward surface, or both, of the image capture device and mirrors a shape of a forward portion or a rearward portion of the image capture device. The planar surface of the heatsink **600** is connected to mounting flanges including a first mounting flange **602A**, a second mounting flange **602B**, and a finger mounting flange **602C** that extend away from the planar surface to support other components.

(78) The mounting flanges **602A**, **602B**, **602C** function to support components, printed circuit boards, electrical elements, batteries, battery cages, or a combination thereof. The mounting flanges **602A**, **602B**, **602C** function to provide a heat transfer surface, ground electrical elements (e.g., a battery or a printed circuit board), protect internal components, or a combination thereof. The mounting flanges **602A**, **602B**, **602C** may create a partial cage around internal components of the image capture device such as a battery, a battery cage, an ISLA, or a combination thereof. The mounting flanges **602A**, **602B**, **602C** may extend normal to a planar surface of the heatsink **600**. The mounting flanges **602A**, **602B**, **602C** may be located in an end corner region of the heatsink **600**. For example, the mounting flanges **602A**, **602B**, **602C** may be located in a top right corner, a top left corner, a bottom left corner, a bottom right corner, or a combination thereof. The mounting flanges **602A**, **602B**, **602C** may be located in a central region of the heatsink **600**. The mounting flanges **602A**, **602B** may be located in an upper region of the heatsink **600** and the mounting flange **602C** may be located in a lower region of the heatsink **600** opposite the upper region. The mounting flanges **602A**, **602B**, **602C** may extend from a location proximate to one or more of the component recesses **604**, **606**, **608**, **610**.

(79) The component recesses **604**, **606**, **608**, **610** function to receive components, partially recess a component, or both. The front side of the heatsink **600** includes the (LCD recess **606** and the microphone membrane recess **608** shown in FIG. **6B** that function to remove thermal mass from the LCD (not shown) and other components in contact with the front side of the heatsink **600**. The rear side of the heatsink **600** includes the microphone recess **610** shown in FIG. **6A** that functions to remove thermal mass from the microphone (not shown) and other components in contact with the rear side of the heatsink **600**.

(80) FIG. **6A** illustrates a shield feature **614** that extends between a first region of the heatsink **600** comprising the microphone recess **610** and the cutout **612** and a second region of the heatsink **600**. As shown, the second region is located on an opposite side of the heatsink **600** as the LCD recess **606** that houses an LCD (not shown). The shield feature **614** may receive a seal or gasket (not shown) that acts as a conductive shield. The shield feature **614** may function to support a seal or gasket that prevents or limits electrical radiation from extending from a first component to a second component or a first region to a second region within the image capture device. For example, the shield feature **614** may retain a seal or gasket that prevents electrical radiation from extending from the LCD to the microphone, the ISLA, or both. The shield feature **614** may prevent some electrical radiation from extending from a region proximate to the shield feature **614**. The shield feature **614** may receive or be a connection location for a gasket or some other member that prevents or limits conduction of electrical radiation. The shield feature **614** may be a raised portion of the heatsink **600**, may create a recess within the heatsink **600** by creating an elevated portion of the heatsink **600**, may be raised so that the seal or gasket is compressed when the image capture device is closed, or a combination thereof.

(81) In another example, the heatsink **600** comprises locator pins **618** that extend away from the heatsink **600** and assist in locating the heatsink **600** within manufacturing equipment, installation equipment, relative to other components of the image capture device, or a combination thereof. The locator pins **618** may function to ensure proper alignment of components relative to each other so that heat is evenly distributed along the heatsink **600**. For example, the locator pins **618** align a main printed circuit board (not shown) with the heatsink **600**.

(82) As illustrated, the heatsink **600** includes battery bosses **620** and connection bosses **622** that connect the heatsink **600** to other components within the image capture device. The battery bosses

620 connect the heatsink **600** to a battery cage (not shown), a battery connector (e.g., the battery connector **522** of FIG. 5), or both, and the connection bosses **622** receive fasteners to connect the heatsink **600** to a housing (not shown), a planar surface of the battery cage (not shown), a bayonet (not shown), or a combination thereof.

(83) FIG. 6B is a front view of the heatsink **600** showing several component recesses **604**. The component recesses **604** may be an absence of material or a reduction in material. For example, the heatsink **600** may be thinner at a location of the component recesses **604**, create a pocket that receives a component of the image capture system, or both. The component recesses **604** may connect to a component and the component may be flush or sub-flush with a plane of the heatsink **600**. The component recesses **604** may reduce an amount that a component (e.g., a microphone, an LCD, or a gasket) extends above a plane of the heatsink **600**.

(84) As shown in FIG. 6B, the heatsink **600** includes a space **616**. The space **616** is located between the LCD recess **606** and the microphone membrane recess **608** so that mass of the heatsink **600** is reduced where no components are present. In the example of FIG. 6B, the space **616** is an absence of material. In other examples, the space **616** may reduce mass of the heatsink **600** or be located in a region with low thermal application. For example, the space **616** may be located in a region where heat is not introduced into the heatsink **600**. The space **616** may be in a location where an opposing component (not shown) extends toward the heatsink so that the space **616** prevents contact between the opposing component and the heatsink **600**. The space **616** may be free of receiving any components. The space **616** may be located on an opposite side or a same side as the mounting flange **602C** (shown in FIG. 6A). The space **616** may be symmetrical; asymmetrical; square; round; oval; geometric; non-geometric; shaped to fit between recesses, cutouts, or components of the image device; or a combination thereof. The space **616** may be located on a front side, a rear side, or both sides of the heatsink **600**. More than one space **616** may be located on the heatsink **600**. For example, a front and a back of the heatsink **600** may each include the space **616**, or the front or the back may include two or more spaces **616**. The space **616** may balance mass in the heatsink **600**, thermal performance of the heatsink **600**, or both. The space **616** may remove mass from a location where there is less of a thermal load so that more mass may be located at an area with a greater thermal load so that a temperature of the heatsink **600** may be substantially equal from side to side or edge to edge (e.g., a left region v. a right region or a top region v. a bottom region).

(85) FIG. 6B illustrates the battery bosses **620** as positioned to allow connection of a battery cage (not shown) to a planar surface of the heatsink **600**. The battery bosses **620** are located adjacent to the cutout **612** so that the battery cage when installed does not interfere with the ISLA (not shown). The heatsink **600** includes mounting holes **624** located throughout the heatsink **600** that may be used to connect the heatsink **600** to other components of the image capture device. The mounting holes **624** may connect the heatsink **600** to a front housing, a battery cage, a microphone, a printed circuit board, an LCD, a bayonet, an ISLA, or a combination thereof.

(86) FIG. 6C is a rear view of the heatsink **600** showing the first mounting flange **602A** and the second mounting flange **602B** that are positioned to support a GPS assembly (not shown). The heatsink **600** includes the finger mounting flange **602C** that connects to and supports a battery cage (not shown) via a battery boss **620**.

(87) FIG. 7A is a top perspective view of a heatsink **700** and a GPS assembly **702** for use with an image capture device such as the image capture devices **100**, **200** of FIGS. 1A-2B. The GPS assembly **702** may function to correlate positioning data with an image or video at a time the image or video is taken by monitoring a position of the image capture device. The heatsink **700** includes mounting flanges **704** including flange bosses **706** that receive fasteners **708** to locate the GPS assembly **702** within an image capture device. The fasteners **708** connect a PCB **710** of the GPS assembly **702** to the flange bosses **706** located in a first mounting flange **712** and a second mounting flange **714**, respectively. The PCB **710** includes an antenna **716** that is located between

the first mounting flange **712** and the second mounting flange **714**. The GPS assembly **702** may include the antenna **716**, the PCB **710**, the heatsink **700**, a switch **720**, a shield (not shown), or a combination thereof. The GPS assembly **702** may be located in an upper portion of the image capture device. The GPS assembly **702** may be located proximate to a housing **718** (FIG. 7B) and the heatsink **700** may orient the GPS assembly **702** relative to the housing **718**. The GPS assembly **702** and the antenna **716** may be located close to the housing **718** without being in contact with the housing **718**. The antenna **716** is located between the PCB **710** and the housing **718**.

(88) FIG. 7B is a cross-sectional view depicting the relationship between the heatsink **700**, the GPS assembly **702**, and the housing **718**. The GPS assembly **702** is connected to the heatsink **700** at the first mounting flange **712** and the second mounting flange **714** by fasteners **708** so that the GPS assembly **702** is fixedly located within an image capture device. The antenna **716** is spaced apart from the housing **718** by a distance that represents a gap G. The gap G between the antenna **716** and the housing **718** may be sized to keep the antenna **716** and the housing **718** from contacting each other. The gap G may be sufficiently small in size such that signals to and from the antenna **716** may enter and exit the housing **718** unimpeded by the housing **718** and other componentry in the image capture device.

(89) For example, the gap G between the housing **718** and the antenna **716** may measure about 0.01 mm or more, 0.05 mm or more, 0.1 mm or more, 0.2 mm or more, or 0.5 mm or more. The gap G between the housing **718** and the antenna **716** may be about 4 mm or less, 2 mm or less, or 1 mm or less. A thickness of the mounting flanges **712**, **714** may affect a size of the gap G. The thickness of the mounting flanges **712**, **714** and the antenna **716** may be substantially the same, the antenna **716** may have a thickness greater than one or both of the mounting flanges **712**, **714**, the antenna **716** may have a thickness less than one or both of the mounting flanges **712**, **714**, or a combination thereof. The antenna **716** may be located on the PCB **710** between the mounting flanges **712**, **714** (e.g., the first mounting flange **712** and the second mounting flange **714**). The PCB **710** of the GPS assembly **702** may be grounded by the mounting flanges **712**, **714**, grounded by the fasteners **708** that connect the GPS assembly **702** to the mounting flanges **712**, **714**, or both. The fasteners **708** may have exposed conductive material (e.g., silver or copper) that may assist in grounding the GPS assembly **702** relative to the mounting flanges **712**, **714**.

(90) FIGS. 8A and 8B are isometric views of a rear side of a heatsink **800**. A conductor **802** is connected to the heatsink **800** and a component of the heatsink (see FIG. 8B) so that the conductor **802** assists in distributing heat about the heatsink **800**. The conductor **802** as shown assists in moving heat from the component that is located on a rear side of the heatsink **800** to a front side of the heatsink **800** so that the heat may be evenly distributed across the heatsink **800** a more readily removed from the heatsink **800**. The conductor **802** may be made of any material that conducts heat. The conductor **802** may be made of a pliable material, a flexible material, a thin material, a sheet of material, graphite, graphene, a metal, copper, gold, silver, nickel, a plated material (e.g., with any of the materials listed herein), or a combination thereof. The conductor **802** may be one monolithic piece of material, a braided material, a pressed material, a woven material, a flattened material, or a combination thereof. The conductor **802** may extend over or around a sheet conductor **804**.

(91) The sheet conductor **804** functions to provide additional conductance between the heatsink **800** and components. The sheet conductor **804** may be made of a same material as the conductor **802**. The sheet conductor **804** may be a graphite sheet. The sheet conductor **804** may be larger than the conductor **802**. The sheet conductor **804** is located proximate to a planar surface **806** of the heatsink **800**. The sheet conductor **804** may be in direct contact with the planar surface **806** so that heat may be distributed to the heatsink **800** through the planar surface **806**. The sheet conductor **804** may be substantially a same size as the planar surface **806**. The sheet conductor **804** may assist in providing a thermal transfer between the heatsink **800** and the component creating heat. The sheet conductor **804** may have a higher thermal conductivity than the heatsink **800**.

(92) The sheet conductor **804** may have a thermal conductivity of about 500 Wm.sup.-1 K.sup.-1 or more, about 700 Wm.sup.-1 K.sup.-1 or more, about 900 Wm.sup.-1 K.sup.-1 or more, about 1200 Wm.sup.-1 K.sup.-1 or more, about 1500 Wm.sup.-1 K.sup.-1 or more, or about 1700 Wm.sup.-1 K.sup.-1 or more (e.g., ± 5). The sheet conductor **804** may have a thermal conductivity from about 25 Wm.sup.-1 K.sup.-1 to about 500 Wm.sup.-1 K.sup.-1, from about 500 Wm.sup.-1 K.sup.-1 to about 2000 Wm.sup.-1 K.sup.-1, from about 750 Wm.sup.-1 K.sup.-1 to about 1800 Wm.sup.-1 K.sup.-1, from about 1000 Wm.sup.-1 K.sup.-1 to about 1500 Wm.sup.-1 K.sup.-1, from about 1100 Wm.sup.-1 K.sup.-1 to about 1300 Wm.sup.-1 K.sup.-1 (e.g., about 1300 Wm.sup.-1 K.sup.-1). The sheet conductor **804** may assist in spreading heat across the heatsink **800**, providing heat into the heatsink **800**, or both. The sheet conductor **804** may assist in pulling heat from one or more heat-producing components and providing the heat into the heatsink **800**.

(93) The planar surface **806** is a large flat area of the heatsink **800**. The planar surface **806** may be a recess, a surface that contacts an LCD (not shown), a surface opposite an LCD, a surface that contacts a battery (not shown), a surface that contacts a battery cage (not shown), a surface that contacts a microprocessor, a surface that contacts a power circuit, a surface that contacts a system on chip, a surface that contacts a power management integrated circuit, or a combination thereof. The planar surface **806** may be the component recess **604**, the LCD recess **606**, or both as shown in FIG. 6B. The planar surface **806** may be similarly sized to the component that the planar surface **806** is receiving a thermal transfer of energy. The planar surface **806** may be larger than the component (e.g., an LCD, power circuit, or microprocessor) that the planar surface **806** is receiving the thermal energy. The planar surface **806** may be sufficiently large to be in direct contact with one or more components, two or more components, or even three or more components. The planar surface **806**, the sheet conductor **804**, or both may be in communication (e.g., in direct contact with) with or bounded by a shield feature **808**.

(94) The shield feature **808** may thermally isolate and/or electrically isolate the planar surface **806**, the sheet conductor **804**, a component in contact with the sheet conductor **804**, or a combination thereof. The shield feature **808** may prevent heat and/or electrical energy from extending from a first side of the shield feature **808** to a second side of the shield feature **808**. The shield **808** may inhibit electrical interference from interfering with an LCD, power circuit, or microprocessor (not shown).

(95) FIG. 8B illustrates the heatsink **800** and the sheet conductor **804** covered by a printed circuit board **810**. The heatsink **800** and the printed circuit board **810** may be in communication (e.g., in direct contact or in indirect contact). For example, a thermal interface material may be located between the heatsink **800** and the printed circuit board **810**. The printed circuit board **810** may include one or more components that generate heat while the image capture device, such as the image capture device **100** or **200** of FIG. 1 or 2, is in use. The components on the printed circuit board **810** may be one or more microprocessors, one or more power circuits, a system on chip, a power management integrated circuit, or a combination thereof. The printed circuit board **810** extends along a first side (e.g., rear side) of the sheet conductor **804** (not shown in FIG. 8B) and the heatsink **800** extends along a second side (e.g., front side) of the sheet conductor **804**. The conductor **802** is connected to the printed circuit board **810** and assists in distributing and/or removing heat from the printed circuit board **810**. The conductor **802** extends from a rear side **812** of the printed circuit board **810** over the printed circuit board **810**, the planar surface **806**, the heatsink **800**, and connects to a front side **814** of the heatsink **800**. The conductor **802** allows heat generated by the printed circuit board **810** or components on the printed circuit board to be moved to front side **814** of the heatsink **800** so that the heatsink **800** dissipates the heat or evenly distributes the heat. The conductor **802** balances heat within the system by moving heat from a location within the system that has a high-level of heat and moving the heat to a location with a lower level of heat. This balancing of heat permits the system to operate longer before a thermal

shut off, operating without a thermal shut off, or both.

(96) FIG. **9** is a front view of a printed circuit board **900**. The printed circuit board **900** can be similar to the printed circuit board **810**. A sheet conductor **902** is in communication with the printed circuit board **900** and/or components on the printed circuit board to assist in removing heat from the printed circuit board **900**. The printed circuit board **900** may include components such as a power circuit **904** and a microprocessor **906**. The power circuit **904** and the microprocessor **906** are located on the printed circuit board **900** and are in communication (e.g., in direct contact with or in indirect contact with) with the sheet conductor **902**. The power circuit **904**, the microprocessor **906**, or both may be in direct contact with the sheet conductor **902**. The sheet conductor **902** provides heat dissipation directly to the power circuit **904** and microprocessor **906** so that heat is rapidly removed from both components. The quick heat dissipation allows the power circuit **904**, the microprocessor **906**, or both to operate without thermal issues or to operate longer without shutting down due to thermal issues.

(97) The power circuit **904** may perform a DC to DC conversion, battery charging, voltage scaling, power sequencing, or a combination thereof. The power circuit **904** may produce heat while the image capture device that includes the printed circuit board **900**, such as the image device **100** or **200** of FIG. **1** or **2**, is being used. The power circuit **904** may be a power management integrated circuit (PMIC). The power circuit **904** may work in conjunction with the microprocessor **906**.

(98) The microprocessor **906** may control the image capture device, store images, store instructions, control one or more sensors, or a combination thereof. The microprocessor **906** may include memory. The microprocessor **906** may be a system on chip (SOC). The microprocessor **906** may produce heat when the microprocessor **906** is operating.

(99) FIG. **10A** is a cross-sectional view of the heatsink **800** of FIG. **8B** along lines XA-XA referring to the heatsink **1000** for clarity. The heatsink **1000** is connected to a printed circuit board **1002**. The printed circuit board **1002** has a microprocessor **1004** that is in communication with a sheet conductor **1006**. The sheet conductor **1006** is sandwiched between a microprocessor **1004** and the heatsink **1000** to facilitate heat transfer from the microprocessor **1004** into the heatsink **1000**. As shown, the sheet conductor **1006** is in indirect contact with the microprocessor **1004** on a first side and the heatsink **1000** on a second side.

(100) The sheet conductor **1006** has a thickness that is less than a thickness of the heatsink **1000**. The sheet conductor **1006** may have a thickness of about 5 μm , about 10 μm or more, or about 15 μm or more. The sheet conductor **1006** may have a thickness of about 1000 μm or less, about 750 μm or less, about 500 μm or less, about 250 μm or less, about 100 μm or less, about 50 μm or less, or about 25 μm or less. The sheet conductor **1006** is connected to the heatsink **1000** by a connector **1007**.

(101) The connector **1007** may be a chemical connection, a mechanical connection, or a combination of both. The connector **1007** may be an adhesive, glue, tape, double sided tape, a conductive adhesive, a sheet of adhesive, a porous adhesive, a porous tape, or a combination thereof. The connector **1007** may extend around a periphery of a planar surface of the heatsink **1000**, the sheet conductor **1006**, or both. The connector **1007** may cover an entire surface of the planar surface of the heatsink **1000**, the sheet conductor **1006**, or both. The connector **1007** may conduct thermal transfer between the sheet conductor **1006** and the heatsink **1000**. The connector **1007** may connect the microprocessor **1004** to the sheet conductor **1006**. The sheet conductor **1006** may be in direct contact but free of physical attachment to the microprocessor **1004**. The sheet conductor **1006** may be in contact with a thermal interface material **1010** and the thermal interface material **1010** may be in indirect or direct contact with the microprocessor **1004**.

(102) FIG. **10B** illustrates the sheet conductor **1006** of FIG. **10A** in direct contact with the heatsink **1000** and a power circuit **1008** located on the printed circuit board **1002**. The sheet conductor **1006** is connected to the heatsink **1000** by the connector **1007** so that contact between the sheet conductor **1006** and heatsink **1000** is maintained. The sheet conductor **1006** facilitates heat transfer

from the power circuit **1008** into the heatsink **1000**. The sheet conductor **1006** may be in direct contact but free of physical attachment to the power circuit **1008**. The heatsink **1000** assists in removing heat, extending operating time, distributing heat, or a combination thereof. The heatsink **1000** may assist in extending a running time of an image capture device such as the image capture device **100** or **200** of FIGS. **1** and **2**, in addition to supporting avoidance of shutdown due to high temperatures. FIG. **10C** is an exploded side view illustrating a stack up of the components. The heatsink **1000** is connected to the sheet conductor **1006** via a connector **1007**. The printed circuit board **1002** is connected to the microprocessor **1004** and the power circuit **1008**. The microprocessor **1004** and the power circuit **1008** both include a thermal interface material **1010**. The thermal interface material **1010** contacts the sheet conductor **1006** so that heat is transferred from the microprocessor **1004** and the power circuit **1008** into the heatsink **1000**.

(103) The thermal interface material **1010** may be any material that assists in heat transfer from the microprocessor **1004**, the power circuit **1008**, or both into the sheet conductor **1006** and the heatsink **1000**. The thermal interface material **1010** may be an adhesive or include an adhesive that connects the thermal interface material **1010** to a component. A connector **1007** may be attached to one or both sides of the thermal interface material **1010** to connect the thermal interface material to components. Thermal interface material **1010** may have a thermal conductivity of about $1 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$ or more, about $5 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$ or more, or about $10 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$ or more. Thermal interface material **1010** may have a thermal conductivity of about $100 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$ or less, about $50 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$ or less, or about $25 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$ or less.

(104) While the disclosure has been described in connection with certain embodiments, it is to be understood that the disclosure is not to be limited to the disclosed embodiments but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. An image capture device comprising: a housing; a battery cage; an integrated sensor and lens assembly; a heatsink located partially or completely within the housing, wherein the heatsink comprises: a cutout through which a portion of the integrated sensor and lens assembly extends; a planar surface located adjacent to the cutout, wherein the planar surface comprises a front side and a rear side; bosses configured to receive fasteners that connect a component to the heatsink; a first mounting flange extending from a top of the heatsink; and a finger mounting flange extending from a bottom of the heatsink; a printed circuit board in communication with the rear side of the planar surface of the heatsink, wherein the printed circuit board and the battery cage are located between the first mounting flange and the finger mounting flange, and wherein the printed circuit board and the heatsink are connected together via one or more of the bosses; and a liquid crystal display (LCD) recess located in the front side of the planar surface of the heatsink that is configured to receive a liquid crystal display (LCD), wherein the LCD recess is opposite the printed circuit board.
2. The image capture device of claim 1, wherein a power circuit, a microprocessor, or both are located on the printed circuit board.
3. The image capture device of claim 1, further comprising: a thermal interface material located between the printed circuit board and the rear side of the planar surface to thermally connect the printed circuit board to the planar surface.
4. The image capture device of claim 3, wherein a thermal conductivity of the thermal interface material is between $1 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$ to $100 \text{ Wm}\cdot\text{sup.}-1 \text{ K}\cdot\text{sup.}-1$.
5. The image capture device of claim 1, wherein the printed circuit board is located between the heatsink and the battery cage.

6. The image capture device of claim 5, wherein the battery cage includes through holes.
7. The image capture device of claim 1, wherein the heatsink is made of aluminum.
8. The image capture device of claim 1, wherein the printed circuit board is located between the heatsink and the battery cage and the printed circuit board is in direct contact with the heatsink.
9. An image capture device comprising: an integrated sensor and lens assembly; a battery cage; a heatsink comprising: a planar surface comprising: a first side facing external to the image capture device, and a second side facing internal to the image capture device; a cutout that is adjacent to the planar surface, wherein a portion of the integrated sensor and lens assembly is located on a first side of the heatsink, the integrated sensor and lens assembly extends through the cutout, and a portion of the integrated sensor and lens assembly is located on a second side of the heatsink; first mounting flanges extending from a top of the heatsink; finger mounting flanges extending from a bottom of the heatsink opposite the first mounting flanges so that the battery cage is extendable between the first mounting flanges and the finger mounting flanges; and bosses configured to receive fasteners that form a connection with the heatsink; and a liquid crystal display (LCD) recess, on the first side of the planar surface, that is configured to connect to a liquid crystal display (LCD) of the image capture device; and a printed circuit board in communication with the second side of the planar surface.
10. The image capture device of claim 9, further comprising: a thermal interface material located between the printed circuit board and a rear side of the planar surface to thermally connect the printed circuit board to the planar surface.
11. The image capture device of claim 10, wherein the printed circuit board is located between the heatsink and the battery cage.
12. The image capture device of claim 9, wherein the cutout is a space through the heatsink that is surrounded by heatsink material.
13. The image capture device of claim 9, wherein the printed circuit board comprises a microprocessor, a power circuit, or both.
14. The image capture device of claim 9, wherein the LCD recess is a reduction in material that creates a pocket to receive a portion of the LCD.
15. An image capture device comprising: a housing; a heatsink located partially or completely within the housing, wherein the heatsink comprises: a planar surface, a cutout configured to receive an integrated sensor and lens assembly, a first mounting flange connected to a first edge of the heatsink, a finger mounting flange connected to a second edge of the heatsink with the planar surface extending between the first mounting flange and the finger mounting flange; and bosses configured to receive fasteners that connect a component to the heatsink; a thermal interface material; a battery cage; a liquid crystal display (LCD) recess, on a first side of the planar surface; and a printed circuit board in communication with the planar surface of the heatsink via the thermal interface material, wherein the battery cage and the printed circuit board are located between the first mounting flange and the finger mounting flange.
16. The image capture device of claim 15, further comprising: a liquid crystal display (LCD) that is in communication with the planar surface on a side of the heatsink opposite the printed circuit board.
17. The image capture device of claim 15, wherein the heatsink is located completely within the housing.
18. The image capture device of claim 15, further comprising: a component recess disposed below the cutout and on an opposite side of the heatsink as the first mounting flange and the finger mounting flange.
19. The image capture device of claim 15, further comprising: a shield feature extending between the printed circuit board and the planar surface of the heatsink.
20. The image capture device of claim 19, wherein the shield feature extends around a periphery of the planar surface.

